
Silver
Ag2+
Ion

Magnesium
Mg2+
Ion

Copper
Cu2+
Ion

Copper
Cu2+
Ion

Carbonatee
CO32-
Ion

Lead
Pb2+
Ion

Lead
Pb2+
Ion

Zinc
Zn2+
Ion

Zinc
Zn2+
Ion

Carbonatee
CO32-
Ion

Barium
Ba2+
Ion

Barium
Ba2+
Ion

Sulphate
SO42-
Ion

Sulphate
SO42-
Ion

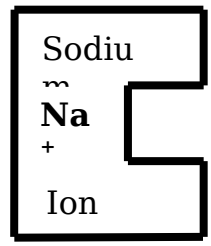
Carbonatee
CO32-
Ion

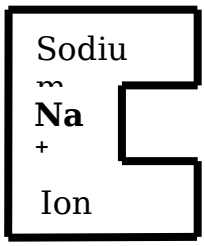
Iron (II)
Fe2+
Ion

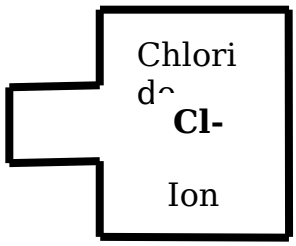
Iron (II)
Fe2+
Ion

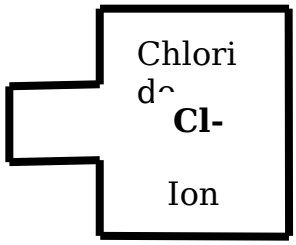
Sulphate
SO42-
Ion

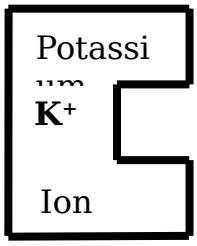
Oxide
O2-
Ion

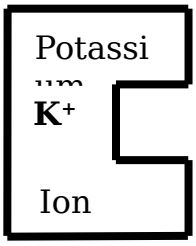

Sodium
Na+
Ion

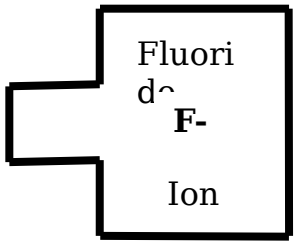

Sodium
Na+
Ion

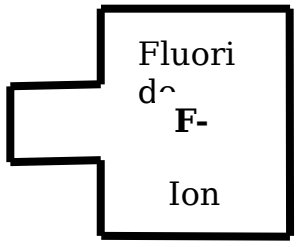

Chloride
Cl-
Ion

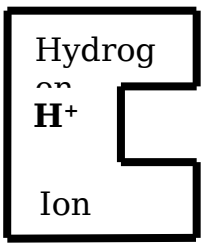

Chloride
Cl-
Ion

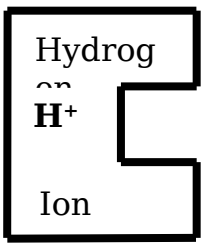

Potassium
K+
Ion

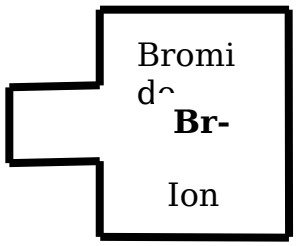

Potassium
K+
Ion

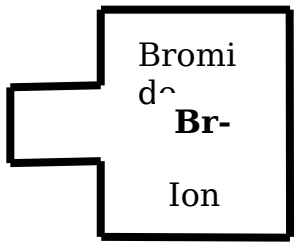

Fluoride
F-
Ion

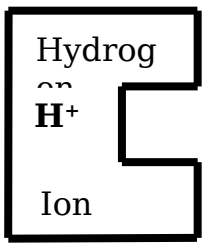

Fluoride
F-
Ion

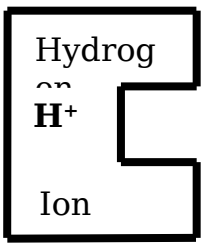

Hydrogen
H+
Ion

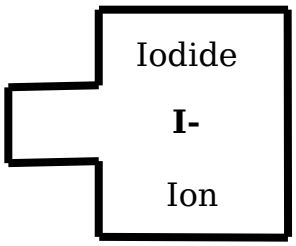

Hydrogen
H+
Ion

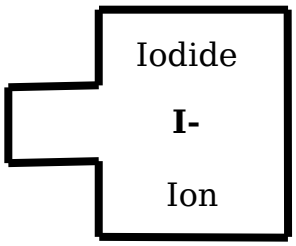

Bromide
Br-
Ion

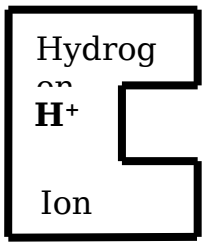

Bromide
Br-
Ion

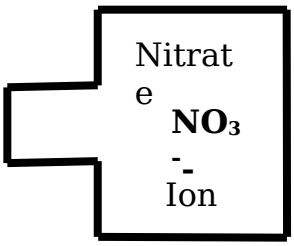

Hydrogen
H+
Ion

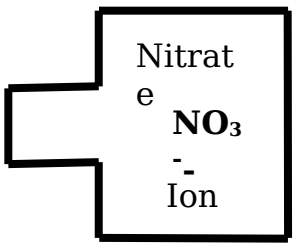

Hydrogen
H+
Ion

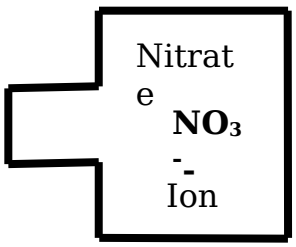

Iodide
I-
Ion

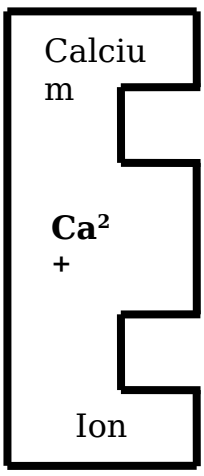

Iodide
I-
Ion

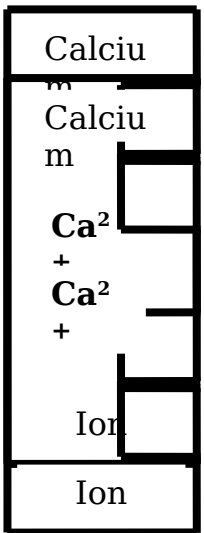

Hydrogen
H+
Ion

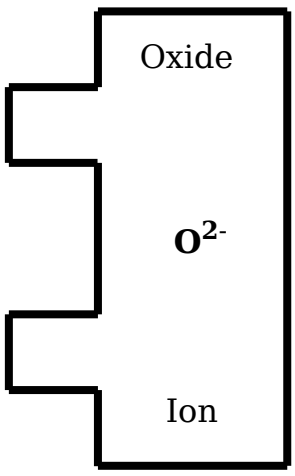

Nitrate
NO3-
Ion

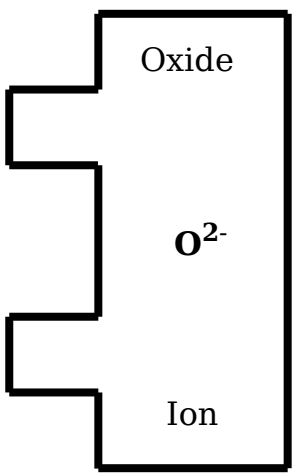

Nitrate
NO3-
Ion

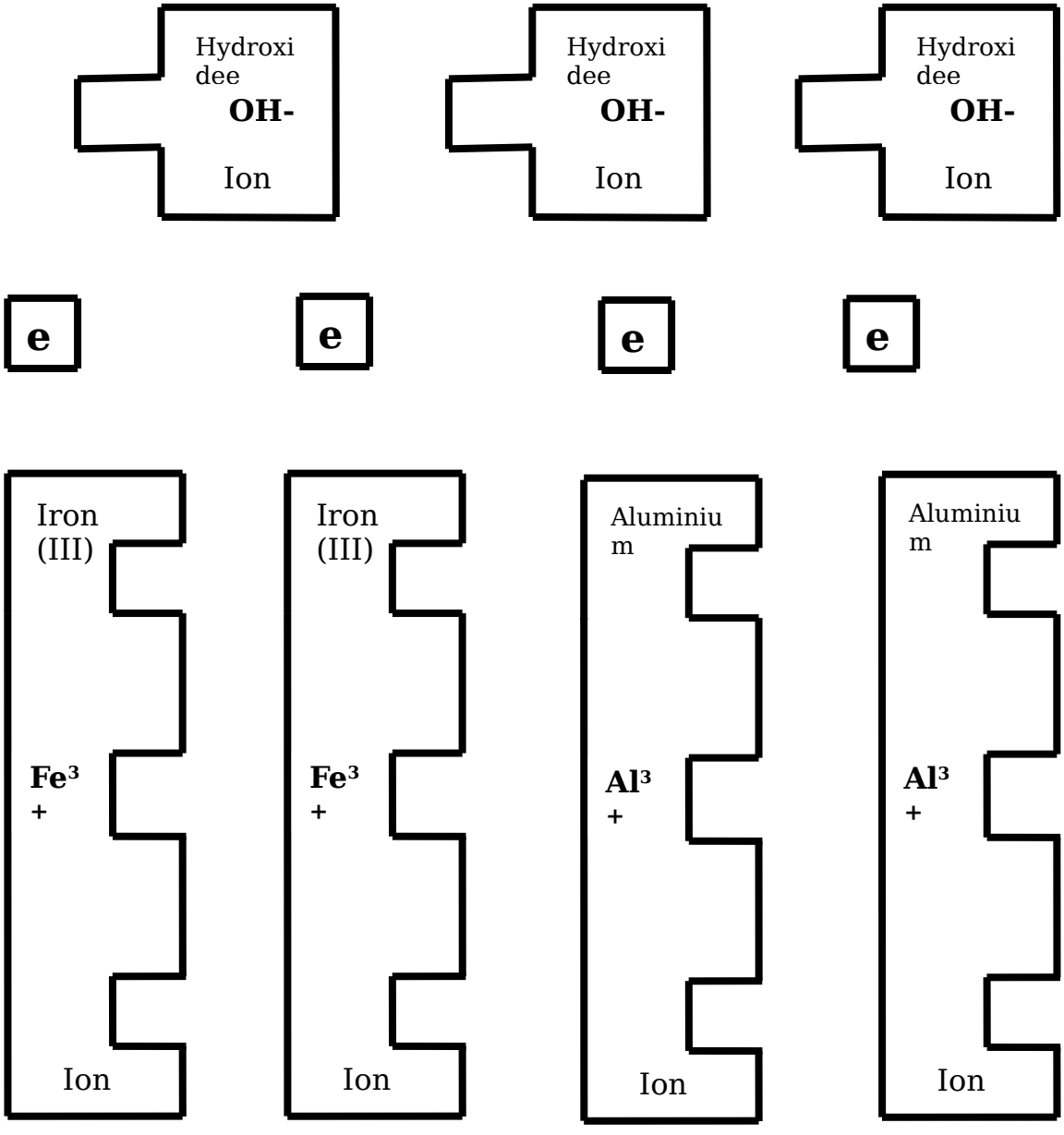

Nitrate
NO3-
Ion


Calcium
Ca2+
Ion


Calcium
Calcium
Ca2+
Ca2+
Ion
Ion


Oxide
O2-
Ion


Oxide
O2-
Ion


Hydroxi
dee
OH-
Ion
Hydroxi
dee
OH-
Ion
Hydroxi
dee
OH-
Ion
e
e
e
e
Iron
(III)
Fe3
+
Ion
Iron
(III)
Fe3
+
Ion
Aluminiu
m
Al3
+
Ion
Aluminiu
m
Al3
+
Ion

Making Ionic Compounds – Using the Ionic jigsaw

Please look after your jigsaw. Try to keep the pieces together in one place on the desk and not mixed up with other sets.

A Tasks

1. **Arrange** your jigsaw into three groups: Metals, Non-metals and Electrons.
2. **Make a list** of the ions in each of the Metals and Non-metals groups for your notes.
3. **Find** a Sodium Ion, a Calcium Ion and an Iron (III) Ion in your jigsaw.
4. **Describe** how you would make each of these Ions into an atom again.
5. Find a Bromide Ion, a Nitrate Ion and an Oxygen Ion.
6. **Describe** how you would make each of these Ions into an atom again.

B Tasks

1. Use the jigsaw to **make** Sodium Chloride (this is the common salt used in cooking).
2. The chemical formula for Sodium Chloride is NaCl. This is because one Sodium Ion fits onto one Chloride Ion with no gaps left. Make a **note** of the name and formula of Sodium Chloride.
3. **Make** and **note** the name and formula for Potassium Chloride, Sodium Iodide and Potassium Fluoride.
4. **Make** and **note** the name and formula for Calcium Oxide, Lead Oxide, Calcium Fluoride and Silver Chloride. (Check with your teacher if you are not sure how to write these down.)

C Tasks

1. **Make** and **note** the name and formula for Zinc Carbonate.
2. Make a **note** of the THREE different atoms that go to make up Zinc Carbonate.
3. **Make** and **note** the name and formula Lead Sulphate, Iron (II) Sulphate, Silver Nitrate and Potassium Nitrate.

4. For each of the four compounds you made in 3 make a **note** of the different atoms that make them up.

D Tasks

1. **Make** and **note** the name and formula for Aluminium Oxide, Iron (III) Oxide and Aluminium Sulphate.
2. **Make** and **note** the name and formula for some of your own chemical compounds.